Week 03 Progress Report

Details

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- **Name of Industry:** SS Infotech Pvt. Ltd.
- **Reporting Week: (19/01/2026 to 23/01/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of a Week

- During Week 3, I focused on understanding essential security fundamentals related to Linux user management, privilege control, authentication logs, and basic system hardening. These tasks were assigned by my industry mentor to help me develop a security-focused mindset required in SOC environments.

- I explored how Linux manages users, groups, and permissions, and analysed which accounts have administrative privileges. I then reviewed authentication logs and monitored how the system records login events, sudo attempts, and user activity.

- Finally, I conducted basic system hardening checks by reviewing password policies, file permissions, running services, and open ports. This helped me understand how secure configurations form the foundation of system defense and how SOC analysts identify deviations during an investigation.

Objectives of Week 3

- Understand Linux user, group, and privilege architecture.
- Learn how authentication and sudo events are logged.
- Establish a baseline for normal system behavior.
- Perform foundational system hardening checks.

Tasks Completed

1. User & Privilege Analysis

- Listed all users using /etc/passwd.
- Listed all groups using /etc/group.
- Identified privileged users in sudo and adm groups.
- Verified personal account privileges using the id command.

2. Authentication Log Analysis

- Viewed recent authentication events in /var/log/auth.log.
- Triggered a harmless sudo access attempt on /root.
- Verified that the event appeared in system logs.
- Filtered sudo-related events using grep to simulate SOC log analysis.

3. Basic System Hardening Checks

- Checked file permissions for /etc/passwd and /etc/shadow.
- Reviewed system password policies from /etc/login.defs.
- Listed all actively running services to identify baseline behavior.
- Performed a fresh open ports baseline using ss -tulnp.

Key Accomplishments

- Developed a strong understanding of Linux identity and privilege structures.
- Learned how authentication and sudo actions are recorded and monitored.
- Improved ability to analyze logs using filtering techniques relevant to SOC investigations.
- Built foundational knowledge of system hardening and secure configuration.
- Established baseline data (users, privileges, logs, services, ports) that will be essential for incident response tasks in upcoming weeks.

Tools & Commands Used

- cat, grep, id, ls -l
- /etc/passwd, /etc/group, /etc/shadow
- /var/log/auth.log, /etc/login.defs
- systemctl, ss, sudo

Learnings / Takeaways

- Privileged accounts must be monitored closely in every system.
- Authentication logs are one of the richest sources for detecting suspicious activity.
- Password policies and file permissions form the foundation of system hardening.
- Running services and open ports determine the system's exposure and attack surface.
- Establishing a security baseline is crucial before moving into SIEM configuration and attack simulation.

Next Week Plan (Week 4)

- Learn SOC fundamentals.
- Begin log analysis exercises.
- Explore basic incident investigation workflows.
- Start preparing for SIEM setup in Week 5.

Screenshots of Activities (if Any)

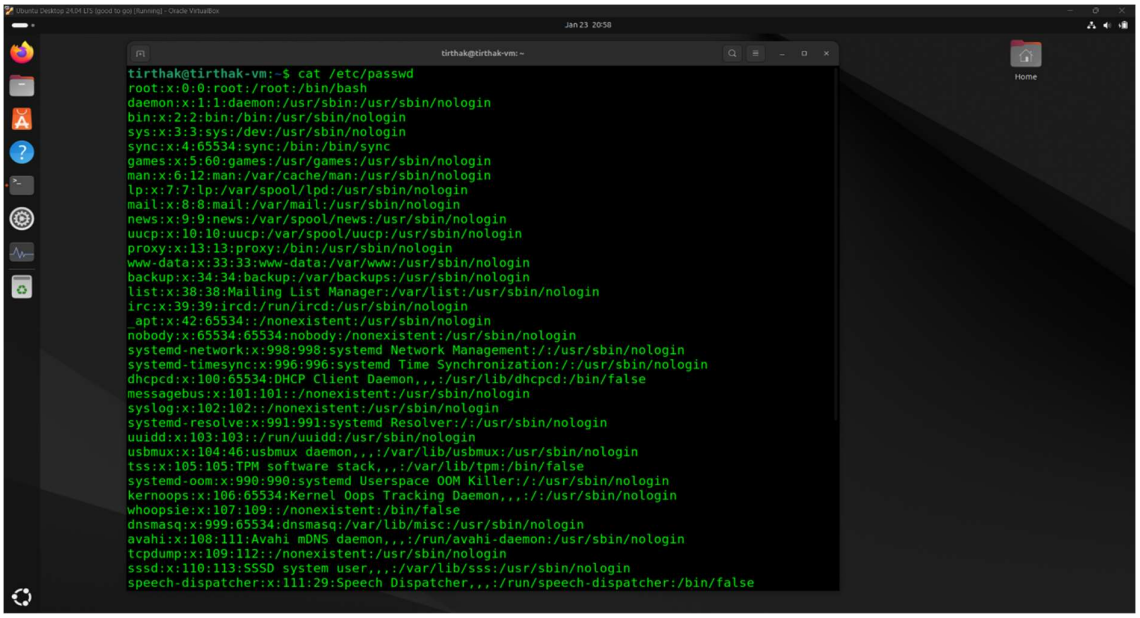


Figure 3.1 – Output of /etc/passwd showing all system users.

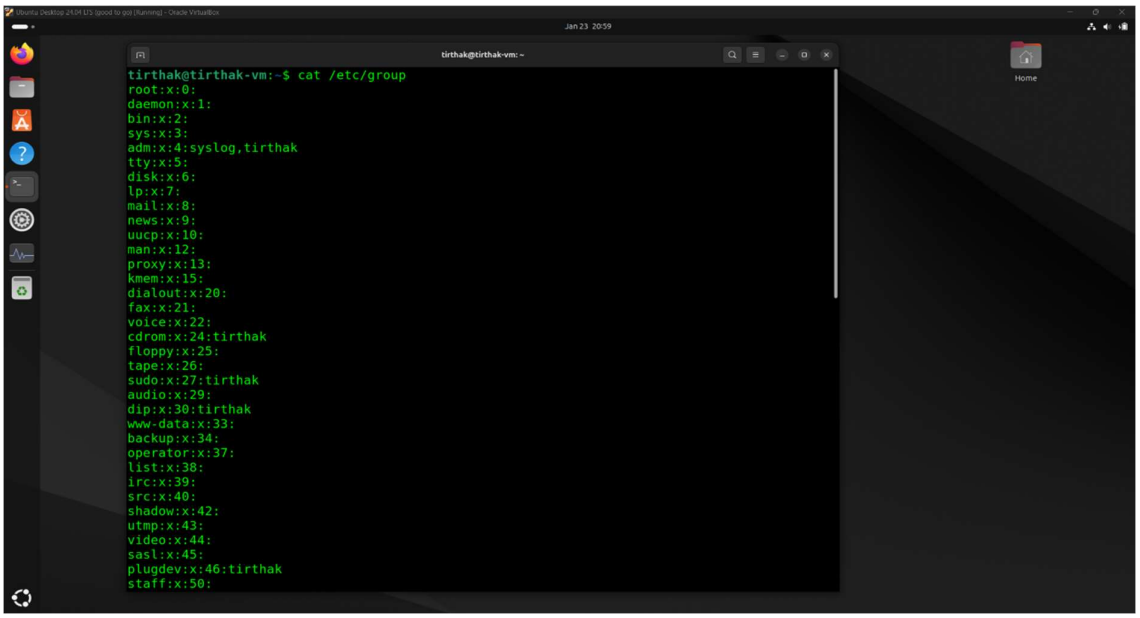


Figure 3.2 – Output of /etc/group showing system groups and memberships.

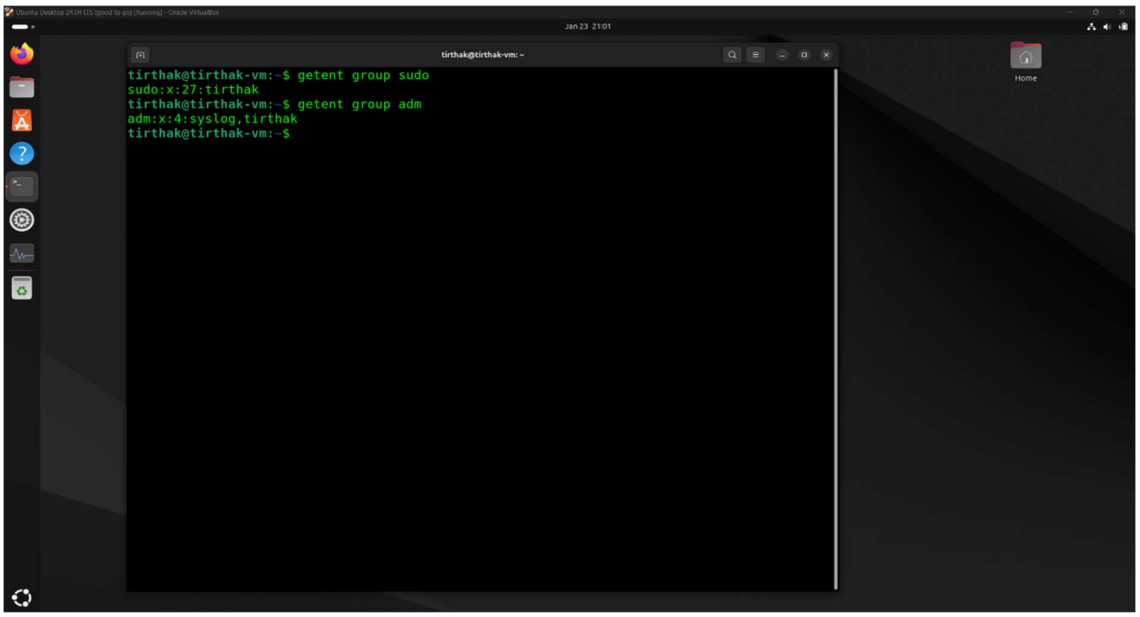


Figure 3.3 – Users with sudo and adm privileges.

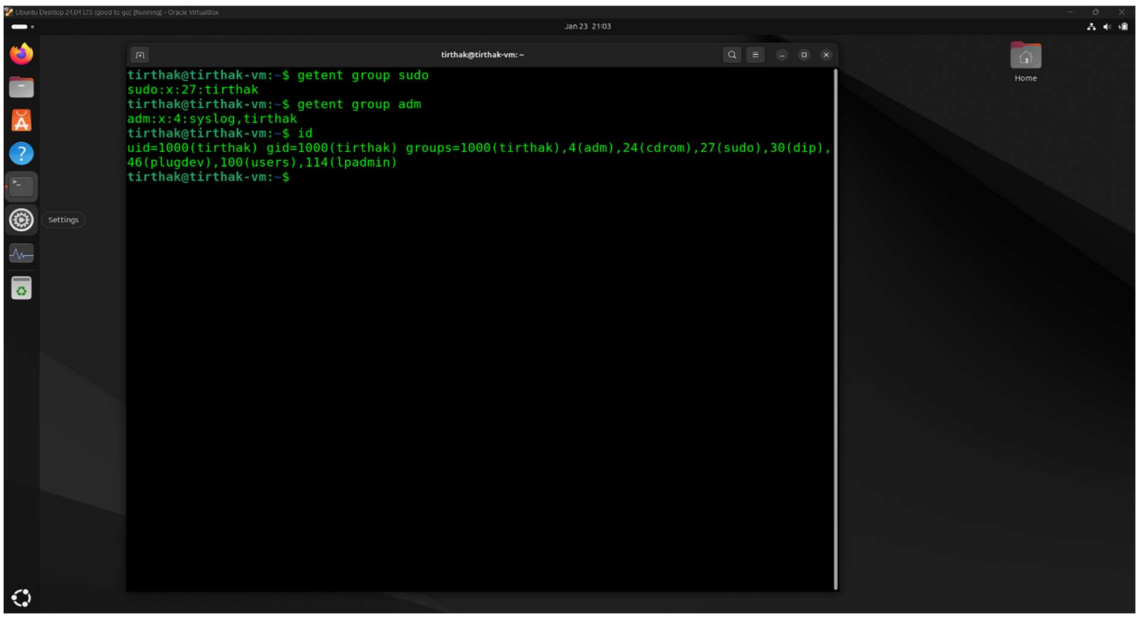


Figure 3.4 – User ID and group membership using id command.

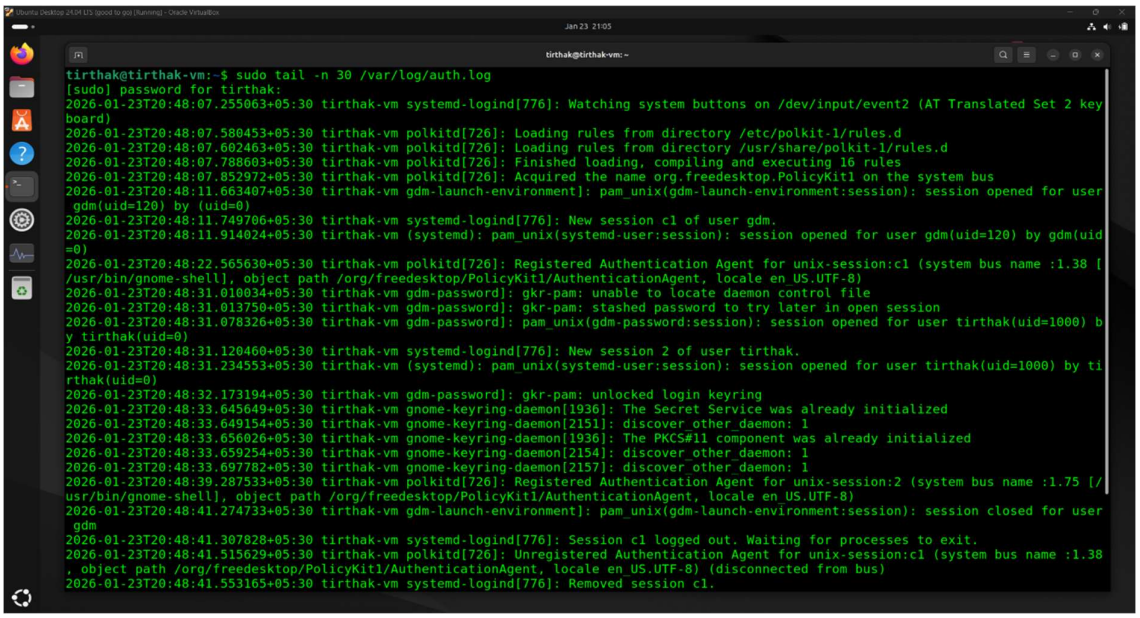


Figure 3.5 – Recent authentication events from /var/log/auth.log.

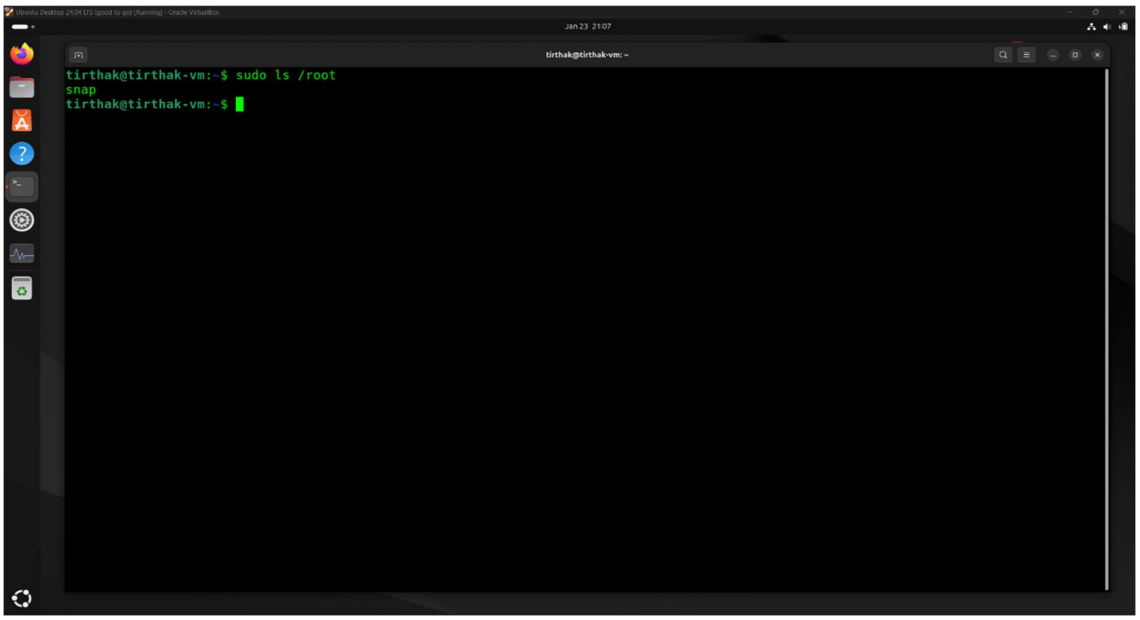
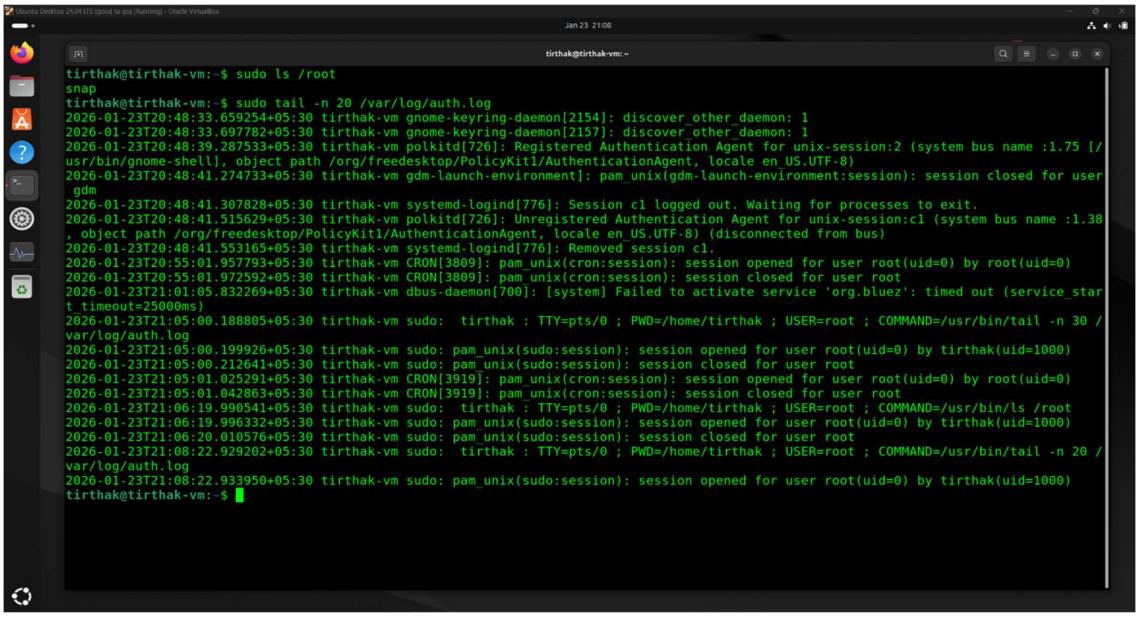


Figure 3.6 – Log entry generated from sudo attempt (ls /root).



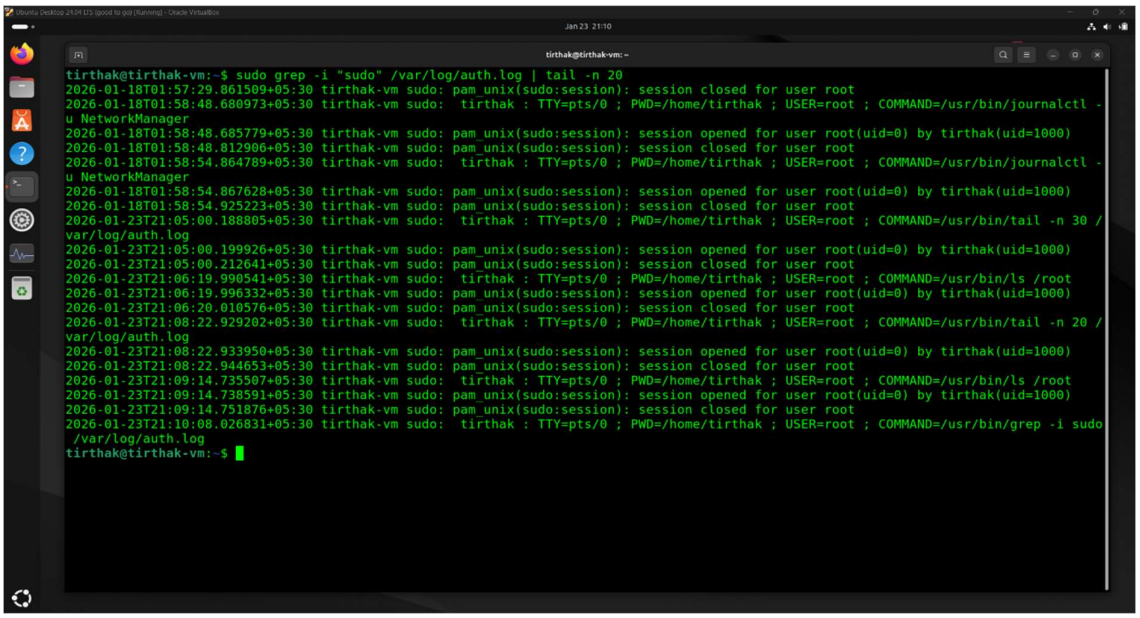


Figure 3.7 – Filtered sudo activity using grep.

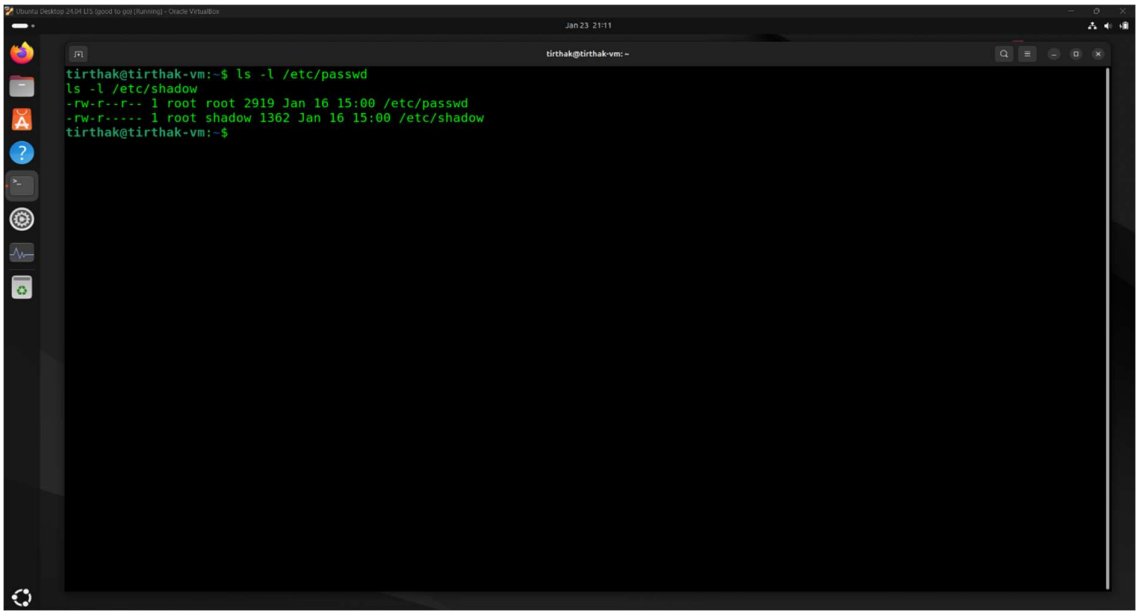


Figure 3.8 – Permissions of /etc/passwd and /etc/shadow files.

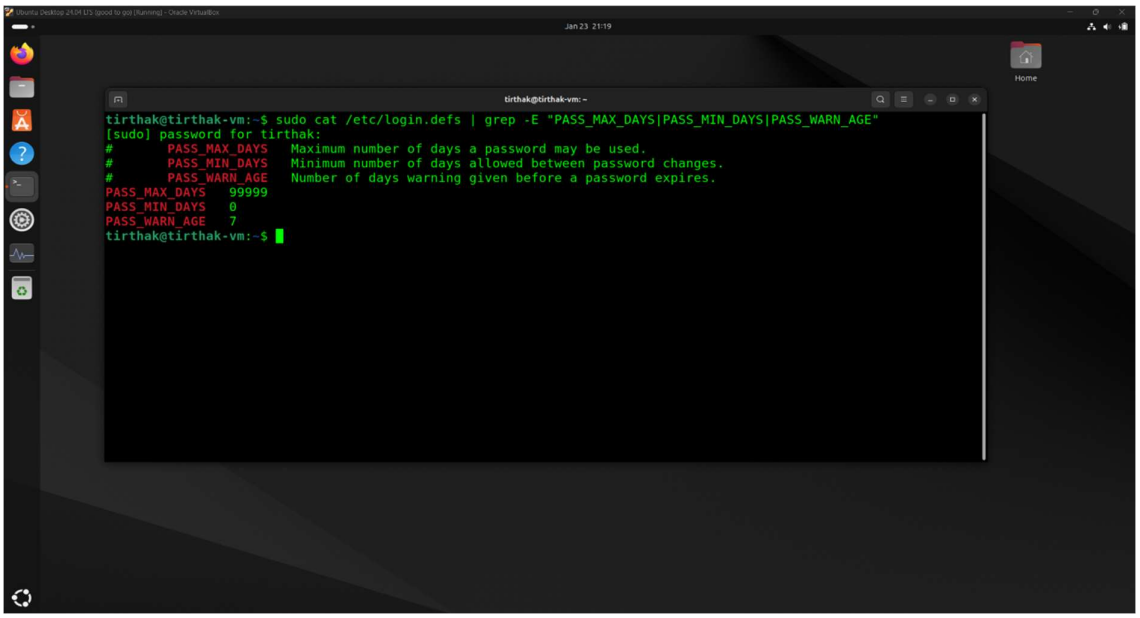


Figure 3.9 – Password policies from /etc/login.defs.

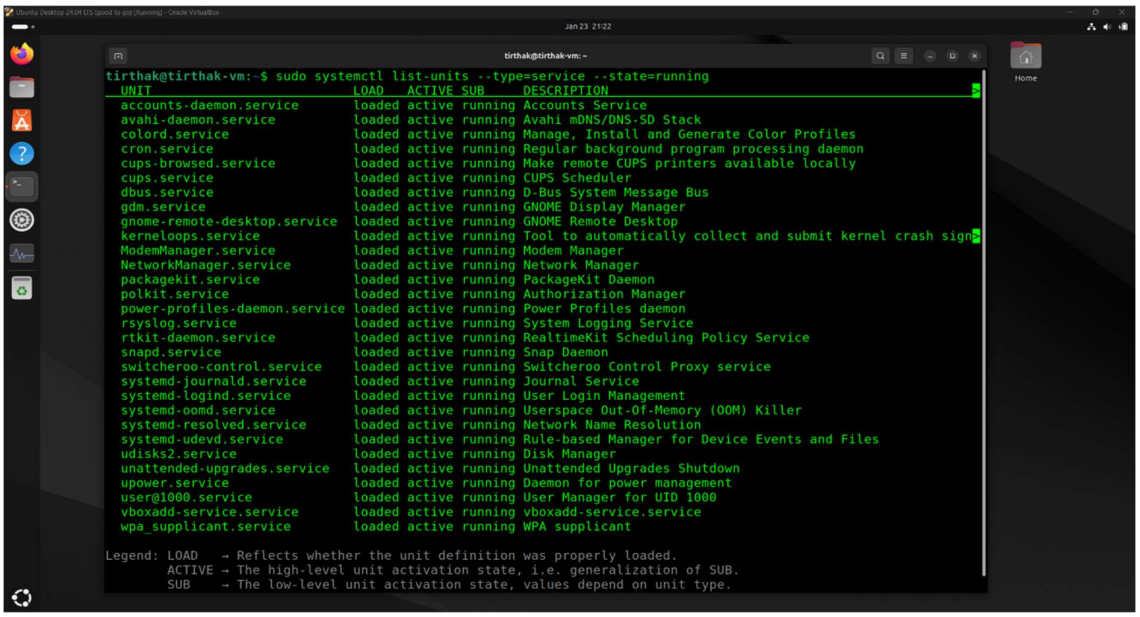


Figure 3.10 – List of currently running services.

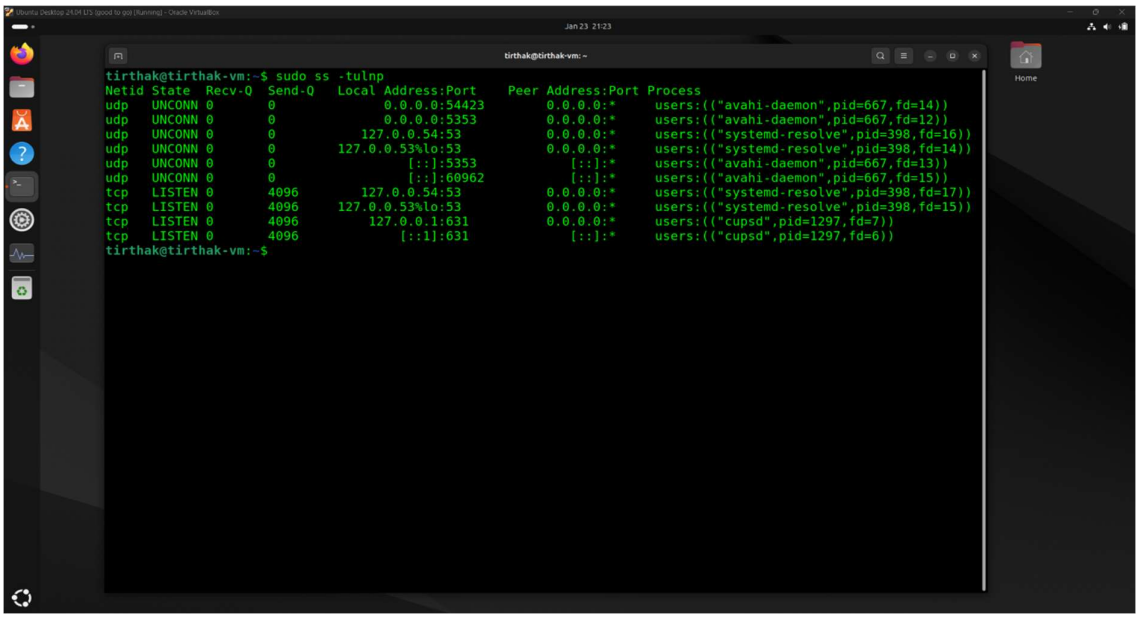


Figure 3.11 – Baseline check of open ports using ss -tulnp.